Urban Passenger Transport

Classification by Type of Usage

- *There are three basic categories of transportation by type of operation and usage: private, for-hire, and public or common carrier*

- *Private transportation consists of privately owned vehicles operated by owners for their own use, usually on publicly provided and operated streets. Motorcycle, bicycle,and, of course, walking also belong in this category.*

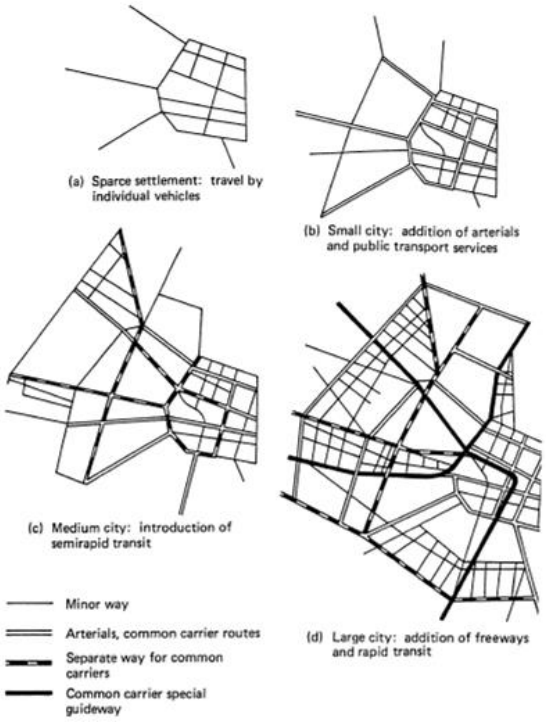
- *For-hire urban passenger transportation, commonly designated as paratransit, is transportation serviceprovided by an operator and available to all parties who meet the conditions of a contract for carriage(i.e., pay prescribed fares or rates), but which is adjustable in various degrees to the individual user's desires.Most paratransit modes do not have fixed routesand schedules. Taxi, dial-a-ride, and jitney are major modes*

- *Public transport, mass transportation, or transit is the common carrier type of urban passenger transport.These are transport systems with fixed routes and schedules, available*

for use by all persons who pay the established fare. The most common representatives are bus, light rail transit, and rapid transit or metro, but there are a number of other modes.

- *Paratransit modes with routes and schedules that change with the requests of individual users are referred to as demand-responsive; when the difference is pointed out, transit is described as fixed-route, fixed schedule service.*

Evolution of a Transportation System in a Model Urban Area



*1 - **Transportation System Evolution with Urban Area Growth***

1. Small Settlements: **Private Travel and Paratransit** - Short trips in the settlement can be performed by walking

(a) Settlement increasing in size - Few dispersed trips would be best served by small motorized vehicles

(b) Further increase in settlement and for improvement of transportation system -Introduction of a common carrier service (taxis)

2. Towns: **Bus Transit and Arterial Streets -** Optimal Vehicle size increases with passenger volume

As passenger volume increases, service with adjustable routing and schedules should be replaced by fixed routes and schedules

3. Medium Sized Cities: **Transit Way Separation and Guided Technology -**

4. Partial separation of modes (transit ROW category B)

5. Introduction of guided transit (LRT)

6. Large Cities: **Construction of Fully Controlled Ways** - Provision of fully controlled ROW for common carriers (rapid transit)

Introduction of transit service results in the following changes:

- Lower-cost transportation becomes available to all persons in the served area
- Simple and conveniently scheduled service is provided throughout the transit network
- Increased street capacity: higher LOS for all vehicles
- Less congestion and its negative impacts

-- Since it is difficult to charge auto drivers the full (including social) cost of their travel, transit mustoften be subsidized to attract its potential passengers.

Replacement of medium by large transit vehicles or units (TUs) results in:

- Higher transporting capacity
- Lower cost per unit capacity, mostly due to higher labor productivity
- Greater riding comfort

-- Lower service frequency for given demand

Construction of higher-capacity roads brings:

- Higher LOS
- Lower unit transportation costs.
- Stimulus to economic growth (result of the preceding two)

-- High investment requirements

-- Negative environmental impact of wide paths and large vehicle storage facilities

Longitudinal separation of transit vehicles (ROW category B) results in:

- Higher level of service and system performance
- Stronger system image and identity
- Higher passenger attraction (consequence of the above two)
- Lower unit operating cost because of higher speed and reliability
- Stronger impact on land use and urban form due to permanence

-- Changed traffic conditions, depending on whether the transit ROW is outside or within existing streets (e.g., curbed median)

-- Additional land area required

-- Substantial investment and construction required

Guided modes, compared with steered transit modes, differ in the following:

- Much larger, spacious, and more comfortable vehicles, particularly with rail technology
- Operation of trains possible, with much greater line capacity and labor productivity
- Lower operating cost per unit of offered capacity
- Electric traction introduced with better performance, no exhaust, and noise
- Greater reliability and safety (fail-safe operation)
- Narrower ROW
- Operation in tunnels, on viaducts, and in park areas feasible without environmental damage

-- Requires higher investment

-- Less compatible with other traffic in street operation

-- Limited to the guideway network only, therefore uneconomical for extensive routing in low-densityareas

-- Lower operational flexibility (rerouting, detours, etc.)

Fully controlled ways (grade-separated roads, ROW A), compared with partially controlled arterials (B) and uncontrolled streets (C), have:

- Much higher performance (capacity, speed, reliability, etc.)
- Higher LOS
- Lower operating costs per vehicle-km
- Greater permanence of system performance (frontages protected from developments)
- Permanence that affects land uses

-- Considerably larger area requirements (particularly for interchanges)

-- Very high investment costs

-- Disruption of corridor during construction

-- A large vehicle volume is attracted into center city

-- Wide highways and large interchanges create physical barriers in the city

Category A, compared with category B ROW, gives transit modes:

- Much higher performance (speed, reliability, capacity due to long trains, etc.)
- Higher LOS
- Lower operating costs per unit capacity due to very high labor productivity
- Stronger image and identity
- Higher passenger attraction (result of the preceding three)
- Stronger positive land-use impacts

- Possibility of automation

-- Need for grade separation of the entire ROW

-- High investment

-- Disruptions of the corridor during the construction

-- Less extensive network (result of the preceding three items)

Fully automated compared with manually operated systems have:

- Increased frequency of service without additional cost
- Lower energy consumption and vehicle wear due to preprogrammed driving
- Easier recovery from service disturbances
- Lower operating cost (if labor savings outweigh the cost of increased system complexity)
- Higher operating safety (human error eliminated)

-- Considerably higher investment cost

-- Much greater technical complexity and therefore lower reliability

-- Requirement for automated track supervision and communication with passengers for handlingemergencies and security

THE FAMILY OF TRANSIT MODES:CATEGORIES AND DESCRIPTIONS

1. **Paratransit**

The paratransit category of transportation systems consistsof a variety of mostly highway vehicles providing services between private cars and fixed-route fixed schedule transit modes. Their types of operation—routing, scheduling and methods of obtaining serviceby users—makes different modes, rather than strictly defined types of vehicles by technology.

Types - Car sharing organizations; Taxis; Dial-a-ride (DAR) and hybrid services; Jitneys

2. **Street Transit Modes**

The most common type of transit in every city, operating on streets, includes three different modes—buses, trolleybuses, and streetcars /tramways.

3. **Medium-Capacity Modes:Semirapid Transit**

Transit modes utilizing mostly ROW category B are characterized by distinctly higher performance—capacity, reliability, speed, and passenger attraction— than the street transit modes.

4. **High-Performance Modes: Rapid Transit**

Exclusive use of ROW category A is the main element required for high performance transit systems. With that ROW, it is logical

that guided technologies, electric traction, train operation, and fail-safe signalization are used, giving the system high capacity, reliability, safety, and other performance elements.

5. **Specialized Transit Modes**

A considerable number of different transportation systems is used as transit modes or components in passenger movements through transit systems.

TRANSIT MODES

A transit mode is defined by its three basic characteristics:

• Right-of-way (ROW) category

• System technology

• Type of service

Rights-of-Way

A transit way or ROW is the travel way or strip of land on which the transit vehicles operate. There are three basic ROW categories distinguished by the degree of their separation from other traffic.

• Category C represents surface streets with mixed traffic. Transit may have preferential treatment,such as reserved lanes separated by lines or special signals or travel mixed with other traffic

- Category B includes ROW types that are longitudinally physically separated by curbs, barriers,grade separation, and the like from other traffic but with grade crossings for vehicles and pedestrians,including regular street intersections. This ROW category is most frequently used for light rail transit (LRT) systems
- Category A is a fully controlled ROW without grade crossings or any legal access by other vehicles or persons. It is also referred to as ''grade separated,"''private,'' or ''exclusive'' ROW, and it can be a tunnel, an aerial structure,or at-grade level

System Technologies

Technology of transit modes refers to the mechanical features of their vehicles and ways. The four most important featuresare defined here-

- **Support** is the vertical contact between vehicle and riding surface, which transfers the vehicle weight and traction force. The most common types are rubber tire on concrete, asphalt, or other surface and steel wheel on steel rail. Other types of support are vehicle body on water (boats and hydrofoils), air cushion (hovercraft), and magnetic levitation
- **Guidance** refers to the means of lateral vehicle guidance. Highway vehicles are steered (by the driver) and their lateral stability is provided by wheel/support adhesion. Rail vehicles are guided by flanges and the conical form of the wheel surfaces. A distinct feature of rail technology is that its wheel/ rail assembly combines both support and guidance
- **Propulsion** refers to the type of propulsion unit and method of traction, or transferring acceleration/ deceleration forces

- **Control** is the means of regulating the travel of one or all vehicles in a system. The most important control is for longitudinal spacing of the vehicles, which may be manual-visual,manual-signal, fully automatic, or various combinations of these

Types of Service

There are many different types of transit services. They can be classified into groups by three characteristics:

First, by the types of routes and trips served:

• Short-haul transit is defined as low- to medium speed services within small areas of high travel density, such as central business districts (CBDs),campuses, airports, and exhibition grounds

• City transit, the most common type, includes transit lines serving an entire city. They may operateon any ROW category (C, B, or A)

• Regional transit consists of long high-speed lines with few stops, serving long trips within the metropolitan region. Regional rail and some express bus lines exemplify this category

Second, by stopping schedule or type of operation:

• Local service is with all transit units stopping at all stops (or as required by passengers)

• Accelerated service is operation when successive TUs skip different sets of stations on a pre-determined schedule (e.g., skip-stop and zonal service)

• Express service is provided when all TUs on a line stop only at widely spaced stops. These lines often parallel local service but serve fewer stops/ stations, comprising express/ local service

Third, by time of operation:

• Regular or all-day service is transit operated duringmost daily hours. This is the basic transit service, and it includes the great majority of transit lines

• Commuter transit or peak-hour service refers to routes operated during peak hours only. They are usually radial from suburbs, focusing on the CBD,and designed for work trips only. Commuter transit is a supplement to an all-day, regular transitbut not a substitute for it.

• Special or irregular service is transit operated only during special events (e.g., sporting events,conventions, exhibitions, or public celebrations)and emergency conditions (snowstorms, floods,etc.).

Generic Classes of Transit Modes

There is no rigorous definition of what differences in ROW,technology, or type of service make a separate mode,but it is common to consider systems as different modes if they differ substantially in one or more of the three characteristics

- Street transit (also known as surface transit) designates modes operated on streets with mixed traffic (ROW category C). Its reliability depends on traffic conditions, primarily traffic congestion and various interferences, and its speed is lower than the speed of traffic flow

due to the time lost at passenger stops; buses, trolley buses, and streetcars /tramways are in this class.

- Semi-rapid transit consists of modes utilizing mostly ROW category B, but C or A may also be used on some sections. This class includes a wide range of modes from those with B and C categories, such as buses and light rail transit(LRT) operating on separated ROW and streets on the low side, to largely grade-separated LRT with tunnels or aerial structures (B and A) on the high side
- Rapid transit modes operate exclusively on category A ROW and have high speed, capacity,reliability, and safety. All existing rapid transitsy stems utilize guided technologies (rail or rubber tire), which permit the operation of trains(high capacity and low operating costs) and automatic signal control (high safety)

Transportation and the Urban Structure

*Urbanization involves an increased numbers of trips in urban areas. Cities have traditionally responded to growth in mobility by **expanding the transportation supply**, by building new highways and transit lines. This has mainly meant building more roads to accommodate an ever-growing number of vehicles. Several **urban spatial structures** have accordingly emerged, with the reliance on the automobile being the most important discriminatory factor. Four major types can be identified at the metropolitan scale:*

- *Type I – Completely Motorized Network: Representing an automobile-dependent city with a limited centrality and dispersed activities.*
- *Type II – Weak Center: Representing the spatial structure of many American cities where many activities are located in the periphery.*
- *Type III – Strong Center: Representing high density urban centers with well developed public transit systems, particularly in Europe and Asia.*
- *Type IV – Traffic Limitation. Representing urban areas that have implemented traffic control and modal preference in their spatial structure. Commonly, the central area is dominated by public transit.*

The Spatial Constraints of Urban Transportation

The amount of urban land allocated to transportation is often correlated with the level of mobility. In the pre-automobile era, about 10% of the urban land was devoted to transportation which was simply roads for a dominantly pedestrian traffic. As the mobility of people and freight increased, a growing share of urban areas was allocated to transport and the infrastructures supporting it. Large variations in the spatial imprint of urban transportation are observed between different cities as well

as between different parts of a city, such as between central and peripheral areas. The major components of the spatial imprint of urban transportation are:

- **Pedestrian areas**. Refer to the amount of space devoted to walking. This space is often shared with roads as sidewalks may use between 10% and 20% of a road's right of way. In central areas, pedestrian areas tend to use a greater share of the right of way and in some instances whole areas are reserved for pedestrians. However, in a motorized context, most pedestrian areas are for servicing people's access to transport modes such as parked automobiles.
- **Roads and parking areas**
- **Cycling areas**
- **Transit systems**
- **Transport terminals**